

INFO6007

Project Management in IT

Week 01: Introduction

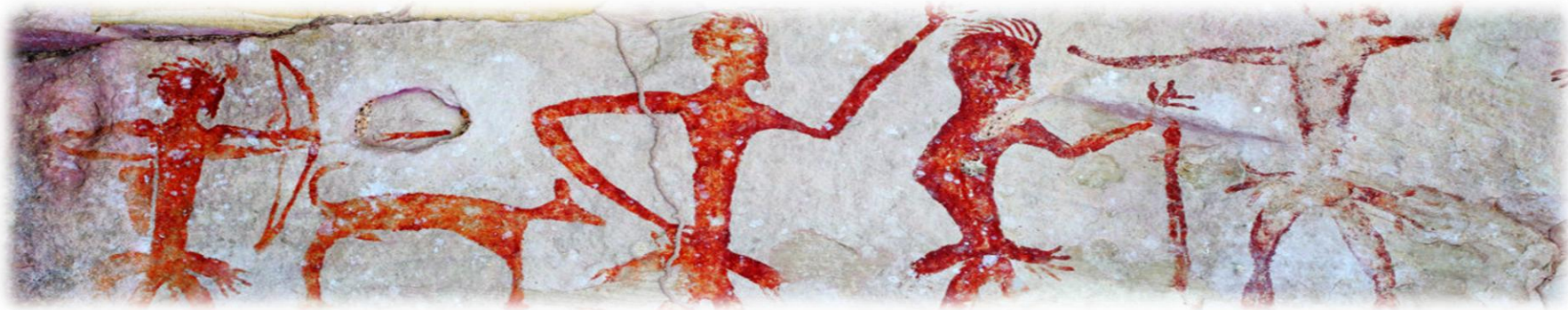
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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Agenda

- UoS overview
- Schedule and expectations
- Assessment Overview

- Introduction
- Project
- Project Portfolio and Program Management
- Project Success
- PM Knowledge Areas

- Words of wisdom from your seniors!

Unit Of Study Overview

Learning outcomes

At the completion of this unit, you should be able to:

LO1. Understand each of the core project management knowledge areas

LO2. Apply project management knowledge, tools and techniques

LO3. Demonstrate developed and refined project management skills

LO4. Develop basic proficiency in project management tools and practices

LO5. Demonstrate team and management skills.

Week	Lecture Topic	Learning Activity	Learning Outcomes
Week 01	Introduction to IT Project Management	Lecture (2 hr)	LO1, LO4
Week 02	IT Project Methods and Requirements Gathering Strategies	Lecture (2 hr)	LO1, LO4
Week 03	Scope and Schedule Management: Tools, Techniques, Real-world Case Study	Lecture (2 hr)	LO1, LO2, LO4
Week 04	IT Project Cost: Tools, Techniques, Real-world Case Study	Lecture (2 hr)	LO1, LO2, LO4
Week 05	IT Project Quality: Tools, Techniques, Real-world Case Study	Lecture (2 hr)	LO1, LO2, LO4
Week 06	IT Project Resources: Tools, Techniques, Real-world Case Study	Lecture (2 hr)	LO1, LO2, LO4
Week 07	IT Project Risk: Tools, Techniques, Real-world Case Study	Lecture (2 hr)	LO2, LO3, LO4
Week 08	Procurement Management: Tools, Techniques, Real-world Case Study	Lecture (2 hr)	LO2, LO4
Week 09	IT Project Leadership, Governance, and Alignment	Lecture (2 hr)	LO1, LO4
Week 10	Integration Management	Lecture (2 hr)	LO1, LO4
Week 11	IT Service Management Functions and Processes 1	Lecture (2 hr)	LO5
Week 12	IT Service Management Functions and Processes 2	Lecture (2 hr)	LO5
Week 13	Course Review	Lecture (2 hr)	LO1, LO4

Topics covered in this unit (Lecture Schedule)

Topics covered in this unit (Tutorial Schedule)

Week	Tutorial Topic	Learning Activity	Learning Outcomes
Week 02	Introduction to IT Project Management	Tutorial (1 hr)	LO1, LO4
Week 03	IT Project Methods and Requirements Gathering Strategies	Tutorial (1 hr)	LO1, LO4
Week 04	Scope and Schedule Management: Tools, Techniques, Real-world Case Study	Tutorial (1 hr)	LO1, LO2, LO4
Week 05	IT Project Cost: Tools, Techniques, Real-world Case Study	Tutorial (1 hr)	LO1, LO2, LO4
Week 06	IT Project Quality: Tools, Techniques, Real-world Case Study	Tutorial (1 hr)	LO1, LO2, LO4
Week 07	IT Project Resources: Tools, Techniques, Real-world Case Study	Tutorial (1 hr)	LO1, LO2, LO4
Week 08	IT Project Risk: Tools, Techniques, Real-world Case Study	Tutorial (1 hr)	LO2, LO3, LO4
Week 09	Procurement Management: Tools, Techniques, Real-world Case Study	Tutorial (1 hr)	LO2, LO4
Week 10	IT Project Leadership, Governance, and Alignment	Tutorial (1 hr)	LO1, LO4
Week 11	Integration Management	Tutorial (1 hr)	LO1, LO4
Week 12	IT Service Management Functions and Processes 1	Tutorial (1 hr)	LO5
Week 13	IT Service Management Functions and Processes 2	Tutorial (1 hr)	LO5

Resources

- Canvas
 - Login using your **Unikey** and **Password**
 - You can access your **lecture recordings, lecture slides, tutorial slides, assignments specs, weekly readings, and announcements** from here.
 - You can also find the **assignment submission links** and **rubrics** here.
- Ed Forum
 - You can **communicate** with the rest of the cohort here.
 - You can ask **doubts to tutors, or the professor via Ed.**
 - **Announcements** would be made on Ed as well!

Weekly Readings

- This unit has its own set of weekly readings prescribed.
- Alongside with the weekly readings, more reading materials will be added to each week's module.
- It is expected that the student will read through the weekly readings and the reading materials on the module before attending the lecture.
- It would be ideal to read the materials before attending the lecture to have a thorough understanding of the unit context.

Language support for International students

- The University of Sydney (USyd) provides various language support services for international students.
- These include:
 - Academic Language and Learning (ALL)
 - Provides support for academic writing, communication, and learning strategies.
 - Students can access resources and workshops to improve their academic performance.
 - <https://www.sydney.edu.au/students/learning-hub/language.html>
 - Speak and Connect
 - A peer-facilitated program for international students to practice English, build confidence, and connect with other students
 - <https://mentoring.sydney.edu.au/p/p7>

Schedule and Expectations

Schedule

- Lecture
 - Session: Every **Tuesday 5 to 7pm**
 - Delivered **Face to Face**, with a **live Zoom option available**.
 - **Recordings available on Canvas** within 24 hours of the end of Session
- Tutorial
 - You **must enrol** in a tutorial
 - Held from **Tuesday to Thursday from Week 2**
 - **Face to Face**
 - Important to attend
 - Please attend the tutorial listed in your timetable

Expectations

- Attend scheduled classes and tutorial.
- Devote about 6-9 hours per week to study apart from lectures and tutorials.
- Read your weekly readings before attending the lecture
- Participate in classes, constructively:
 - Respect for one another
 - Be humble! None of us know everything, but everyone knows something!
- Check canvas regularly - at least once a week!
- Check Ed-Forum regularly - at least once a day!
- Notify academics whenever there are difficulties.
- Notify your team members honestly and promptly about difficulties.

Consultations

- **Fortnightly consultations** will be held for the students from week 2. Announcement will be made on Ed with date and zoom link
- You could use this consultation hour **to discuss doubts regarding the course, assignments**, etc. For which you would like to understand directly from the teaching team rather than from Ed.

Assessment Overview

Assessments

Type	Description	Weight	Due	Length	Use of AI
Written exam 🚧	Final Exam Closed-book supervised exam. One A4 sheet of handwritten and/or typed notes double-sided allowed.	50%	Formal exam period	2 hours	AI prohibited
Outcomes assessed: LO1 LO2 LO3 LO4					
In-person practical, skills, or performance task or test	Oral/Viva based evaluation Viva will be conducted during tutorial session from week 8. It will be based on the topics discussed in the first 7 weeks lectures/tutorials. More detail will be provided on the canvas website.	10%	Multiple weeks	10 minutes	AI prohibited
Outcomes assessed: LO1 LO2 LO3					
Out-of-class quiz 🗨️	Early Semester Feedback Task This is an individual, auto-graded quiz that includes multiple choice questions. It will be based on the contents covered from Week 1 to Week 3.	5%	Week 04	n/a	AI allowed
Outcomes assessed: LO1 LO2					
Written work 🐾	Group project This is a major group project that will be developed incrementally throughout the semester, focusing on understanding how projects are planned, executed, and controlled within organisations to implement technology successfully.	35%	Week 12	n/a	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4 LO5					
🚧 = hurdle task ? 🐾 = group assignment ? 🗨️ = early feedback task ?					

Assessments

- **Early Semester Feedback Task (5%)**
 - Multiple choice, True/False
 - On Canvas – Open book
 - Syllabus – Week 1 to 3
 - **Held in Week 4**
- **Viva / Interactive Oral (10%)**
 - Viva Conducted during the tutorials
 - **Secured assessment** – No notes allowed
 - Syllabus – **Week 1 to Week 7**
 - **Held in Week 8, 9 and 10**

Assessments

- **Group Assignment (35%)**
 - Groups of **4–5 students**: Please form your group **within the same tutorial activity**
 - **Submission in Week 12**
- **Final Exam (50%)**
 - Restricted open-book exam: One A4 sheet allowed (double-sided; handwritten or typed)
 - Case Study based Questions
 - **Held in Exam week.**

Hurdle Requirement

- Students must achieve **at least 40% in the written examination** and obtain **an overall final mark of 50 or more to pass the unit.**
- Any student who does not meet these hurdles requirements may be awarded a maximum final mark of 45, regardless of their overall average.

Best Participation Certificates and Best Project Certificates

- **Best Participation Certificates:** Based on active participation in tutorials; tutors will select best participants from each activity (number of recipients may vary).
- **Best Project Certificates:** Awarded to groups demonstrating outstanding project quality, strong application of concepts, and effective teamwork
- **Purpose:** To recognise engagement, collaboration, and excellence in learning

Guest Lecturer

- Guest lecture delivered by industry professionals
- Opportunity to gain practical, **real-world perspectives beyond the classroom**
- In a previous offering of this unit, a student was offered a **summer internship** through a guest lecture, which later progressed into a **part time role**.
- Session details will be announced on Ed Discussion, **please keep an eye on ED**

Benefits of This Unit: Project Management in IT

- **Develop practical, industry-relevant IT project management skills**
- **Learn to plan, manage, and deliver real-world IT projects** with confidence
- **Strengthen employability and career readiness** through applied learning and industry insights
- **Build capabilities aligned** with roles such as Project Manager, Business Analyst, and IT Consultant
- **Achieve outcomes** that are clearly mapped to the unit learning outcomes, ensuring relevance and progression

A quick 5 min stretch



Project Management in IT

INFO6007

Introduction

- What is a Project?
 - A project is “a temporary effort/work undertaken to create a unique product, service, or result. [(PMBOK® Guide) – Sixth Edition (2017), p. 4.]
- Key characteristics of a Project
 - Temporary
 - It has a clear start and end point.
 - It is not an ongoing operation.
 - Unique outcome
 - The goal is to produce something new, whether it is a product, service, event, etc.
 - Specific objectives
 - Projects aim to meet particular goals, requirements, or solve a problem.
 - Constraints
 - Projects work within limitations of scope, time, cost and resources.

Examples of IT Project



Software Development



Cybersecurity projects



Infrastructure upgrade



System Integration



Data & Analytics



Website/Applications

Why IT Projects Are Inherently Difficult

IT projects operate in environments of uncertainty, change, and hidden complexity.

- Evolving requirements
- Integration complexity
- Decisions under incomplete information

As a result, successful IT projects depend not only on technical skill, but on **professional judgement and the ability to adapt.**

Why IT Projects Fail Despite Good Technology

While IT projects are inherently complex, many failures occur not because of complexity itself, but because of how we manage that complexity.

Across industries, many IT projects struggle due to recurring patterns:

- Poorly defined or changing requirements
- Underestimated complexity and dependencies
- Unrealistic cost and schedule expectations
- Misalignment between business and technical teams
- Weak governance and stakeholder engagement

These issues are **rarely caused by technology alone.**

What Does Project Management Do?

- Project management is the structured process of planning, organising, leading, and controlling resources to achieve specific project objectives within defined constraints.
- “The application of knowledge, skills, tools, and techniques to project activities to meet project requirements.”[(PMBOK® Guide) – Sixth Edition (2017), p. 10.]
- Key elements of Project Management are:
 - Scope - What is the project supposed to achieve? What are its deliverables?
 - Time - How long will it take? What are the deadlines?
 - Cost - What budget is available? What resources are needed?
 - Quality - Are the outcomes meeting the required standards?
 - Risk - What could go wrong, and how do we mitigate it?
 - Communication - How will information be shared among stakeholders?
 - Resources - What people, tools, and technologies are needed?

Why is IT Project Management critical?

- IT projects are often **complex**, **high-stakes**, and directly tied to an organization's **strategic goals**.
 - Key reasons:
 - Aligns technology with Business goals
 - Delivers value on time and within budget
 - Manages Complexity across teams
 - Avoids scope creep and missed deadlines
 - Reduces risks and improves accountability
 - Improves Communication
 - Handles change effectively
-



Image source: <http://www.aussieitgroup.com.au/blog/web-designer-project-management-tools/>

What happens without IT Project Management?

- Scope Creep
- Miscommunication
- Missed deadlines
- Budget blowouts
- Technical Failures
- Stakeholder dissatisfaction



Skills required for Project Management

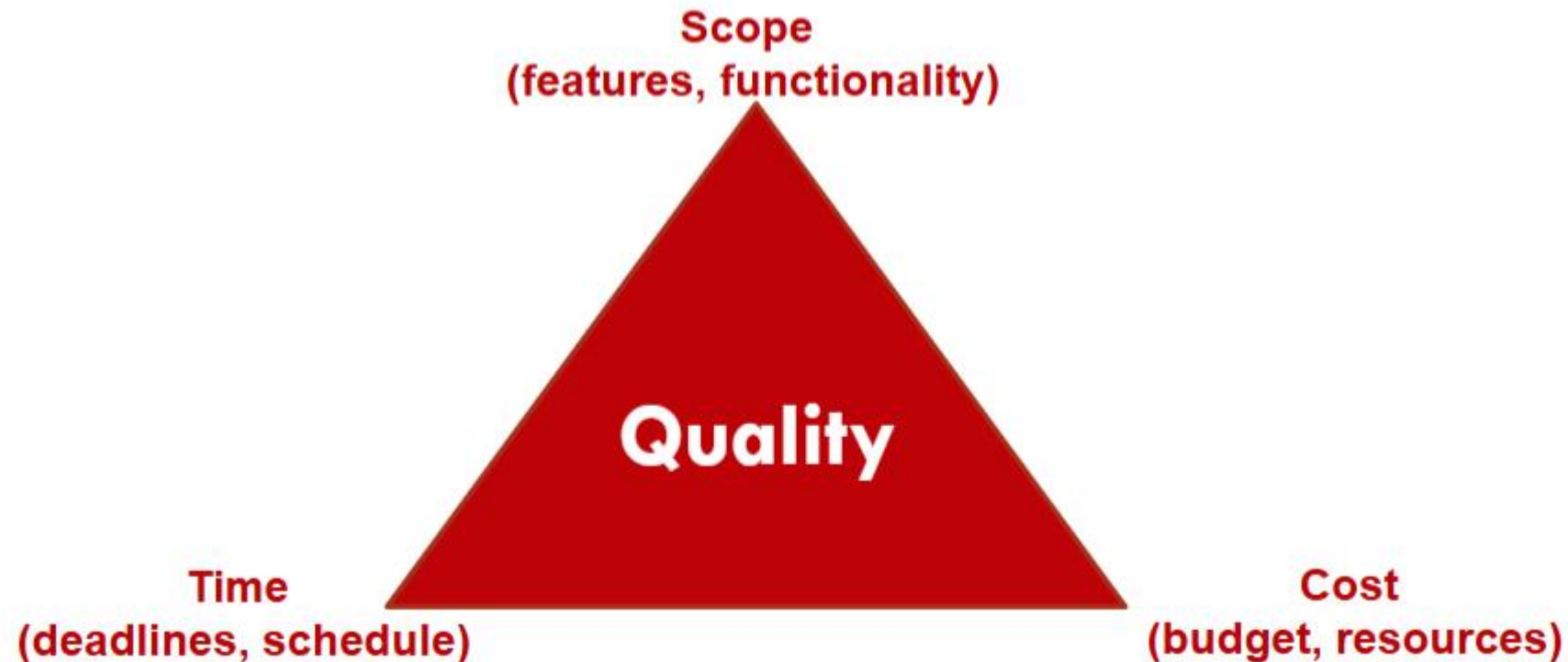
- Core PM skills
 - Planning & Scheduling, Scope management, Risk management, Budgeting & cost control, Quality management and Documentation and Reporting.
- Soft skills
 - Communication, Leadership, Conflict resolution, Negotiation, Decision making, and Adaptability
- Technical skills
 - Understanding of PM methodologies, Tools like Jira, Trello, or Asana, Reporting, etc.
- Strategic skills
 - Stakeholder management, Business Acumen, and Change management
- BONUS TRAITS!
 - Detail oriented but Big picture thinkers, Calm under pressure
 - Proactive problem solvers, Empathetic Team Leaders

Triple Constraints of Project Management

- The **Triple Constraints** refer to the **three key factors** that define and shape any project.
- **Scope**
 - The work required to complete the project.
 - Includes features, functions, tasks, and deliverables.
 - Changes in scope (also called scope creep) can affect time and cost.
- **Time**
 - The schedule or duration of the project.
 - Includes deadlines, milestones, and timelines.
 - Delays often mean increased costs or reduced scope.
- **Cost**
 - The budget for completing the project.
 - Includes resources, labor, tools, software, and infrastructure.
 - Going over budget may require reducing scope or extending deadlines.

Triple Constraints of Project Management contd.

If Scope increases	Time and/or Cost must also increase
If Time is reduced	Cost must increase or Scope must decrease
If Budget is reduced	Scope must reduce



Project Success

- Project Success refers to how well a project meets or exceeds its defined objectives, expectations, and stakeholder satisfaction across several key dimensions.
- Success is no longer just about "on time, on budget", it's also about delivering value.
- Key dimensions of Project success:

Time	Cost	Scope	Quality
Stakeholder Satisfaction	Business Value	Team Performance	Sustainability
User adoption and satisfaction	Alignment with strategic goals	Long term operational impact	Risk management and flexibility

Common misconceptions about success

- “Finishing on time and budget = success”
 - Not if the product isn’t usable or valuable.
- “Meeting business goals is enough”
 - Ignoring team morale or stakeholder input may cause long-term damage.
- “All success is immediate”
 - Some projects show impact only in the long term (especially in IT or transformation projects).

The progression of Project Management

The 3 tiers

- There are 3 levels to management.
- Project Management → Program Management → Portfolio Management
- These three tiers represent **increasing levels of strategic oversight** and coordination in managing work across an organization.

Project Management – The Tactical Execution

- Focuses on **delivering a specific output** (product/service).
- Managed by a **Project Manager**.
- Follows defined **scope, time, and budget** (triple constraints).
- Key objective: **Deliver a successful project**.
- Example: Launching a new website, or developing a mobile app

Program Management – The Strategic Coordination

- A **program** is a collection of **related projects** managed in a **coordinated way** to achieve **benefits** not possible by managing them separately.
- Overseen by a **Program Manager**.
- Focuses on **interdependencies, risks, and strategic alignment**.
- Key objective: **Deliver business value** across multiple projects.
- Examples: Digital transformation program or a HealthTech system rollout

Portfolio Management – The Enterprise Alignment

- A **portfolio** is a **collection of programs and projects** grouped to meet **strategic business goals**.
- Managed by a **Portfolio Manager** or PMO (Project Management Office).
- Focuses on **investment decisions, resource allocation, and strategic prioritization**.
- Key objective: **Maximize value and alignment with organizational strategy**.
- Examples:
 - IT Portfolio: Includes cybersecurity, cloud, AI, and productivity tools
 - Product Portfolio: All R&D and market launch efforts across products

	Projects	Programs	Portfolios
Definition	A project is a temporary endeavor undertaken to create a unique product, service, or result.	A program is a group of related projects, subsidiary programs, and program activities that are managed in a coordinated way to obtain benefits not available from managing them individually.	A portfolio is a collection of projects, programs, subsidiary portfolios, and operations managed as a group to achieve strategic objectives.
Management	Project managers manage the project team to meet the project objectives.	Programs are managed by program managers who ensure that program benefits are delivered as expected, by coordinating the activities of a program's components.	Portfolio managers may manage or coordinate portfolio management staff, or program and project staff that may have reporting responsibilities into the aggregate portfolio.
Monitoring	Project managers monitor and control the work of producing the products, services, or results that the project was undertaken to produce.	Program managers monitor the progress of program components to ensure that the overall goals, schedules, budget, and benefits of the program are met.	Portfolio managers monitor strategic changes and aggregate resource allocation, performance results, and risk of the portfolio.
Success	Success is measured by product and project quality, timeliness, budget compliance, and degree of customer satisfaction.	A program's success is measured by the program's ability to deliver its intended benefits to an organization, and by the program's efficiency and effectiveness in delivering those benefits.	Success is measured in terms of the aggregate investment performance and benefit realization of the portfolio.

A comparative overview of Projects, Programs, and Portfolios

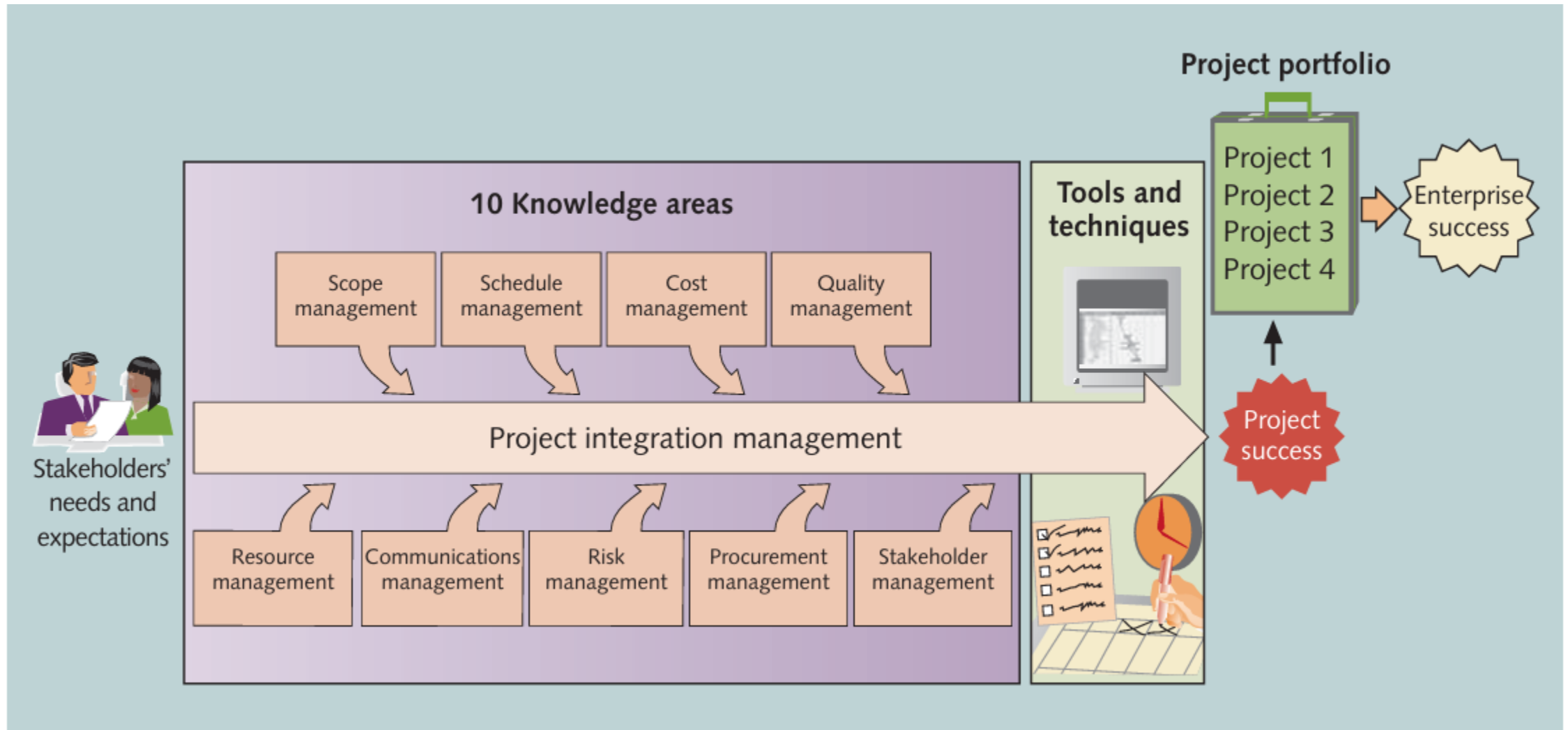
Project Management Framework

Introduction

- A **Project Management Framework** is a **structured approach** that defines **how** projects are planned, executed, monitored, and completed. It provides **guidelines, processes, tools, and best practices** to ensure successful project delivery.
- Core components of a Project Management Framework are:
 - Project Lifecycle
 - Processes and Knowledge Areas
 - Tools and Templates
 - Methodologies within the framework

Why use a Framework?

- Ensures **consistency** across projects
- Reduces **risk** and increases **predictability**
- Improves **communication and accountability**
- Supports **decision-making** with structured processes
- Aligns with **organizational goals and standards**



Source: Project Management Institute, Inc., *A Guide to the Project Management Body of Knowledge (PMBOK® Guide) – Sixth Edition* (2017).

Time for a Menti Break!

- Let's pause for a quick interactive session
- Go to www.menti.com
- Enter the code shown on the screen
- Use your phone, tablet, or laptop
- Participate in quizzes, polls, or share your thoughts

Get ready to reflect, engage, and have fun!

What's Next?

In the coming weeks, we'll explore key components of a Project Management Framework including lifecycle, processes, tools, and methodologies to support effective project delivery.

References

- Watt, A. (n.d.). Project Management. 2nd ed. Victoria B.C.
- Schwalbe, K. (n.d.). Information Technology Project Management. 9th ed. Cengage.

For You!

End of Lecture Questions

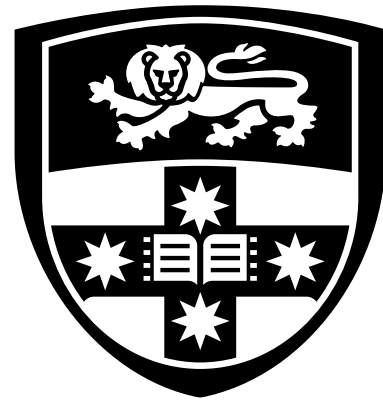
- What are the five phases of a project lifecycle? Can you briefly explain each?
- How do scope, time, and cost interact in the Triple Constraint?
- What is the difference between a project, a program, and a portfolio?
- Name three tools commonly used in IT project management and their purpose.
- If a stakeholder requests a major change late in the project, what steps should you take?

Case Study

- Aurora Tech launched an internal IT project to build a custom CRM system for its sales department. The project had a 6-month timeline and a \$500,000 budget. It involved software development, integration with legacy systems, and user training.
- What went wrong?
 - Scope creep: Sales leadership requested multiple new features during execution without formal change management.
 - Poor communication: Developers were not aligned with end-user needs, leading to usability issues.
 - Lack of stakeholder involvement: Sales team was not engaged during testing, resulting in rework.
 - Underestimated technical complexity: Integration with legacy systems caused repeated delays.
- Outcome
 - Project extended by 4 months and exceeded the budget by 40%.
 - CRM system adoption was low, and trust in the IT department declined.

Case study based Questions

- What were the main causes of project failure in this case?
- How could better project management practices have prevented these issues?
- Which tools or frameworks would you apply to turn this project around?



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